

CRISP-DM Analysis Report

CRISP-DM Studio

Dataset Analysis Report

284807 source rows • 31 source columns • 283726 rows after structural preparation

Executive summary

The uploaded dataset contains 284,807 rows and 31 columns.

The schema contains 31 numeric, 0 categorical/boolean, 0 datetime-like, and 0 identifier-like column(s) based on observed values and names.

No business objective, target definition, decision cost, or domain context was supplied in the CSV itself.

Time has 25,000 observed numeric values; median 8.486×10^4 ; skewness -0.04; 95th percentile 1.639×10^5 (7.908×10^4 above the median).

V1 has 25,000 observed numeric values; median 0.02386; left skew -3.52; 95th percentile 2.079 (2.055 above the median).

V1 has 610 observations (2.4%) outside the descriptive 1.5*IQR fences [-4.24, 4.652]. These are diagnostic flags only; Phase 4 does not remove or clip them.

Clustering: Best silhouette was 0.080, below the meaningfulness threshold 0.20; separation is too weak to present as segments.

HistGradientBoosting is the strongest defensible candidate: held-out $r^2=0.724$, improving on baseline by +0.724.

Business Understanding

KNOWN facts

The uploaded dataset contains 284,807 rows and 31 columns.

The schema contains 31 numeric, 0 categorical/boolean, 0 datetime-like, and 0 identifier-like column(s) based on observed values and names.

No business objective, target definition, decision cost, or domain context was supplied in the CSV itself.

INFERRED analytical objectives

Assess data quality and prepare a defensible dataset for exploratory analysis.

Explore distributions, relationships, and natural segments where the available feature types support them.

Evaluate supervised learning only after a user confirms an appropriate target; current plausible candidates include Class, Amount, Time.

The inferred objectives are analytical hypotheses generated from dataset structure, not claims about the uploader's real business intent.

Data Understanding

284807 Rows

31 Columns

0 Missing cells

1081 Duplicate rows

Per-column profile

Column Inferred type Missing Unique Observed profile

Time numeric 0.0% 124592 0.0% missing; 124,592 distinct observed values; most common observed value share 0.0%; median 8.469e+04; skewness -0.0356.

V1 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.01811; skewness -3.28.

V2 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.06549; skewness -4.62.

V3 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.1798; skewness -2.24.

V4 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.01985; skewness 0.676.

V5 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.05434; skewness -2.43.

V6 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.2742; skewness 1.83.

V7 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.0401; skewness 2.55.

V8 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.02236; skewness -8.52.

V9 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.05143; skewness 0.555.

V10 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.09292; skewness 1.19.

V11 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.03276; skewness 0.357.

V12 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.14; skewness -2.28.

V13 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.01357; skewness 0.0652.

V14 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.0506; skewness -2.

V15 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.04807; skewness -0.308.

V16 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.06641; skewness -1.1.

V17 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.06568; skewness -3.84.

V18 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.003636; skewness -0.26.

V19 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.003735; skewness 0.109.

V20 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.06248; skewness -2.04.

V21 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.02945; skewness 3.59.

V22 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.006782; skewness -0.213.

V23 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.01119; skewness -5.88.

V24 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.04098; skewness -0.552.

V25 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.01659; skewness -0.416.

V26 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median -0.05214; skewness 0.577.

V27 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.001342; skewness -1.17.

V28 numeric 0.0% 275663 0.0% missing; 275,663 distinct observed values; most common observed value share 0.0%; median 0.01124; skewness 11.2.

Amount numeric 0.0% 32767 0.0% missing; 32,767 distinct observed values; most common observed value share 4.8%; median 22; skewness 17.

Class numeric 0.0% 2 0.0% missing; 2 distinct observed values; most common observed value share 99.8%; median 0; skewness 24.

Identifiers and leakage-review flags

No identifier-like columns were detected.

No leakage-review flag was emitted by the structural heuristics.

Candidate supervised-learning targets

Column Task Score Reasons Caveats

Class classification 0.850 Its name resembles a commonly used outcome/label field. It is numeric but has only 2 distinct values, so classification may be plausible. It has no missing target values. It is the final column, a weak structural signal sometimes used for labels.

Amount regression 0.450 It is numeric with enough variation for a regression target. It has no missing target values.

Time regression 0.450 It is numeric with enough variation for a regression target. It has no missing target values.

V1 regression 0.450 It is numeric with enough variation for a regression target. It has no missing target values.

V10 regression 0.450 It is numeric with enough variation for a regression target. It has no missing target values.

V11 regression 0.450 It is numeric with enough variation for a regression target. It has no missing target values.

V12 regression 0.450 It is numeric with enough variation for a regression target. It has no missing target values.

V13 regression 0.450 It is numeric with enough variation for a regression target. It has no missing target values.

Data-quality warnings

1,081 duplicate row(s) were detected (0.4% of rows).

'Class' is near-constant: its most common observed value represents 99.8% of non-missing rows.

Data Preparation

284807 Rows before

283726 Rows after structural

31 Columns before

31 Columns after structural

Decision Category Why Rows affected Columns affected Before ? after

Remove exact duplicate rows applied_now_structural Exact duplicate records add repeated weight without adding information; removal uses no learned statistics. 1081 0 284,807 rows ? 283,726 rows

Remove constant columns applied_now_structural Constant columns contain no observed variation and therefore cannot discriminate records. This structural decision does not estimate a model parameter. 0 0 31 columns ? 31 columns

Exclude identifier-like columns applied_now_structural Identifier-like columns usually represent record identity rather than reusable signal and can enable memorization. They are removed before generic modeling unless a later domain-specific override is introduced. 0 0 31 columns ? 31 columns

Repair losslessly numeric string columns applied_now_structural A string column is converted only when every observed non-missing value parses as numeric; no distributional threshold or fitted statistic is used and no new missing values are introduced. 0 0 ? ? ?

Defer missing-value imputation deferred_to_modeling Numeric median imputation is required for 0 missing numeric cell(s); categorical most-frequent imputation is required for 0 missing categorical cell(s). Medians/modes are learned parameters, so they must be fit on training folds only. 0 0 ? ? ?

Defer categorical encoding deferred_to_modeling One-hot encoding preserves nominal categories. Rare/high-cardinality levels are grouped by OneHotEncoder using training-fold frequencies only, with unknown levels handled safely at transform time. Category vocabularies/frequencies must be learned from training folds only. 0 0 ? ? ?

Handle high-cardinality categoricals inside encoder deferred_to_modeling High-cardinality categoricals are retained but OneHotEncoder will cap/group levels based on training-fold frequencies, preventing a vocabulary learned from validation/test data. 0 0 ? ? ?

Defer numeric outlier clipping deferred_to_modeling Numeric values are clipped to training-fold IQR fences rather than deleting rows. The fences must be learned inside each training fold to avoid leakage. 0 31: Time, V1, V2, V3, V4, V5, V6, V7, V8, V9, V10, V11, V12, V13, V14, V15, V16, V17, V18, V19, V20, V21, V22, V23, V24, V25, V26, V27, V28, Amount, Class ? ? ?

Defer numeric scaling deferred_to_modeling StandardScaler means and standard deviations are learned parameters. Scaling is therefore represented inside the sklearn pipeline and will be fit on training folds only. 0 31: Time, V1, V2, V3, V4, V5, V6, V7, V8, V9, V10, V11, V12,

V13, V14, V15, V16, V17, V18, V19, V20, V21, V22, V23, V24, V25, V26, V27, V28, Amount,
Class ? ? ?

Exploratory Data Analysis

Time has 25,000 observed numeric values; median 8.486e+04; skewness -0.04; 95th percentile 1.639e+05 (7.908e+04 above the median).

V1 has 25,000 observed numeric values; median 0.02386; left skew -3.52; 95th percentile 2.079 (2.055 above the median).

V1 has 610 observations (2.4%) outside the descriptive 1.5×QR fences [-4.24, 4.652]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V2 has 25,000 observed numeric values; median 0.05688; left skew -4.49; 95th percentile 1.754 (1.697 above the median).

V2 has 1,181 observations (4.7%) outside the descriptive 1.5×QR fences [-2.673, 2.868]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V3 has 25,000 observed numeric values; median 0.1772; left skew -1.67; 95th percentile 2.043 (1.866 above the median).

V3 has 292 observations (1.2%) outside the descriptive 1.5×QR fences [-3.743, 3.896]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V4 has 25,000 observed numeric values; median -0.01699; skewness 0.63; 95th percentile 2.568 (2.585 above the median).

V4 has 962 observations (3.8%) outside the descriptive 1.5×QR fences [-3.218, 3.107]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V5 has 25,000 observed numeric values; median -0.0551; skewness 0.42; 95th percentile 2.087 (2.142 above the median).

V5 has 1,129 observations (4.5%) outside the descriptive 1.5×QR fences [-2.632, 2.535]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V6 has 25,000 observed numeric values; median -0.2747; right skew 1.05; 95th percentile 3.143 (3.417 above the median).

V6 has 2,012 observations (8.0%) outside the descriptive 1.5×QR fences [-2.506, 2.129]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V7 has 25,000 observed numeric values; median 0.04373; skewness -0.64; 95th percentile 1.378 (1.335 above the median).

V7 has 747 observations (3.0%) outside the descriptive 1.5×QR fences [-2.235, 2.248]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V8 has 25,000 observed numeric values; median 0.02327; left skew -8.45; 95th percentile 1.035 (1.011 above the median).

V8 has 2,103 observations (8.4%) outside the descriptive 1.5×QR fences [-1.01, 1.124]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V9 has 25,000 observed numeric values; median -0.0521; skewness 0.56; 95th percentile 1.757 (1.809 above the median).

V9 has 697 observations (2.8%) outside the descriptive 1.5×QR fences [-2.504, 2.447]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V10 has 25,000 observed numeric values; median -0.09379; right skew 1.70; 95th percentile 1.534 (1.627 above the median).

V10 has 831 observations (3.3%) outside the descriptive 1.5×QR fences [-2.002, 1.912]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V11 has 25,000 observed numeric values; median -0.04203; skewness 0.35; 95th percentile 1.61 (1.652 above the median).

V11 has 54 observations (0.2%) outside the descriptive 1.5×QR fences [-3.026, 2.999]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V12 has 25,000 observed numeric values; median 0.1376; left skew -2.20; 95th percentile 1.239 (1.102 above the median).

V12 has 1,330 observations (5.3%) outside the descriptive 1.5×QR fences [-1.92, 2.149]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V13 has 25,000 observed numeric values; median -0.01309; skewness 0.07; 95th percentile 1.613 (1.626 above the median).

V13 has 261 observations (1.0%) outside the descriptive 1.5×QR fences [-2.64, 2.663]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V14 has 25,000 observed numeric values; median 0.04883; left skew -1.90; 95th percentile 1.378 (1.329 above the median).

V14 has 1,218 observations (4.9%) outside the descriptive 1.5×QR fences [-1.804, 1.858]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V15 has 25,000 observed numeric values; median 0.05006; skewness -0.32; 95th percentile 1.355 (1.305 above the median).

V15 has 281 observations (1.1%) outside the descriptive 1.5×QR fences [-2.393, 2.463]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V16 has 25,000 observed numeric values; median 0.07367; left skew -1.18; 95th percentile 1.34 (1.266 above the median).

V16 has 694 observations (2.8%) outside the descriptive 1.5×QR fences [-1.942, 2.009]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V17 has 25,000 observed numeric values; median -0.06351; left skew -4.07; 95th percentile 1.253 (1.316 above the median).

V17 has 593 observations (2.4%) outside the descriptive 1.5*IQR fences [-1.785, 1.71]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V18 has 25,000 observed numeric values; median -0.005621; skewness -0.19; 95th percentile 1.412 (1.418 above the median).

V18 has 624 observations (2.5%) outside the descriptive 1.5*IQR fences [-2.02, 2.015]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V19 has 25,000 observed numeric values; median -0.001883; skewness 0.15; 95th percentile 1.28 (1.282 above the median).

V19 has 892 observations (3.6%) outside the descriptive 1.5*IQR fences [-1.822, 1.823]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V20 has 25,000 observed numeric values; median -0.06343; left skew -3.80; 95th percentile 0.8116 (0.875 above the median).

V20 has 2,364 observations (9.5%) outside the descriptive 1.5*IQR fences [-0.725, 0.6445]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V21 has 25,000 observed numeric values; median -0.02805; right skew 5.01; 95th percentile 0.54 (0.568 above the median).

V21 has 1,265 observations (5.1%) outside the descriptive 1.5*IQR fences [-0.859, 0.8234]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V22 has 25,000 observed numeric values; median 0.01037; skewness -0.25; 95th percentile 1.134 (1.124 above the median).

V22 has 116 observations (0.5%) outside the descriptive 1.5*IQR fences [-2.164, 2.17]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V23 has 25,000 observed numeric values; median -0.01191; left skew -5.26; 95th percentile 0.4835 (0.4954 above the median).

V23 has 1,601 observations (6.4%) outside the descriptive 1.5*IQR fences [-0.6236, 0.6083]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V24 has 25,000 observed numeric values; median 0.04177; skewness -0.57; 95th percentile 0.8643 (0.8225 above the median).

V24 has 428 observations (1.7%) outside the descriptive 1.5*IQR fences [-1.555, 1.643]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V25 has 25,000 observed numeric values; median 0.01345; skewness -0.32; 95th percentile 0.7523 (0.7389 above the median).

V25 has 493 observations (2.0%) outside the descriptive 1.5*IQR fences [-1.307, 1.339].

These are diagnostic flags only; Phase 4 does not remove or clip them.

V26 has 25,000 observed numeric values; median -0.05517; skewness 0.56; 95th percentile 0.9292 (0.9843 above the median).

V26 has 520 observations (2.1%) outside the descriptive 1.5×QR fences [-1.172, 1.088]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V27 has 25,000 observed numeric values; median 0.0002727; left skew -2.29; 95th percentile 0.3839 (0.3836 above the median).

V27 has 3,379 observations (13.5%) outside the descriptive 1.5×QR fences [-0.3153, 0.3351]. These are diagnostic flags only; Phase 4 does not remove or clip them.

V28 has 25,000 observed numeric values; median 0.01144; right skew 9.31; 95th percentile 0.2569 (0.2454 above the median).

V28 has 2,651 observations (10.6%) outside the descriptive 1.5×QR fences [-0.2482, 0.2736]. These are diagnostic flags only; Phase 4 does not remove or clip them.

Amount has 25,000 observed numeric values; median 22.4; right skew 9.50; 95th percentile 363 (340.6 above the median).

Amount has 2,699 observations (10.8%) outside the descriptive 1.5×QR fences [-102.6, 186.2]. These are diagnostic flags only; Phase 4 does not remove or clip them.

Class has 25,000 observed numeric values; median 0; right skew 25.26; 95th percentile 0 (0 above the median).

Class has 39 observations (0.2%) outside the descriptive 1.5×QR fences [0, 0]. These are diagnostic flags only; Phase 4 does not remove or clip them.

Time and V3 have Pearson $r=-0.44$ across 25,000 complete observations, indicating a negative linear association.

Analyses skipped by design

Categorical distributions: No categorical or boolean columns are present.

Missingness visualization: No missing cells remain after structural cleaning.

Numeric ? categorical relationships: At least one usable numeric and one usable categorical column are required.

Categorical associations: At least two usable categorical columns are required.

Distribution of Time

Distribution of Time

Outlier diagnostic for Time

Outlier diagnostic for Time

Distribution of V1

Distribution of V1

Outlier diagnostic for V1

Outlier diagnostic for V1

Distribution of V2

Distribution of V2

Outlier diagnostic for V2

Outlier diagnostic for V2

Distribution of V3

Distribution of V3

Outlier diagnostic for V3

Outlier diagnostic for V3

Distribution of V4

Distribution of V4

Outlier diagnostic for V4

Outlier diagnostic for V4

Distribution of V5

Distribution of V5

Outlier diagnostic for V5

Outlier diagnostic for V5

Numeric correlation matrix

Numeric correlation matrix

Clustering

Clustering skipped. Best silhouette was 0.080, below the meaningfulness threshold 0.20; separation is too weak to present as segments.

Silhouette by candidate k

Silhouette by candidate k

K-Means inertia by candidate k

K-Means inertia by candidate k

Predictive Modeling

Split first, fit second: every model is a single sklearn Pipeline containing a fresh Phase 3 preprocessor plus the estimator. Imputation, IQR bounds, scaling, and encoding are learned from training data only; CV clones and refits the whole pipeline inside each fold. The original Phase 3 PreprocessingSpec is never fitted or mutated.

Confirmed target: Time . Detected task: regression . 'Time' is numeric with 124592 distinct values, so it is routed to regression.

Training rows: 16000; held-out test rows: 4000; stratified split: False; CV folds: 3.

Model Comparison

Model Baseline Test metrics CV mean – std Divergence

Dummy baseline True mae=42212.317; rmse=46786.896; r2=-0.000 mae=42838.830 – 77.435; rmse=47495.870 – 111.258; r2=-0.000 – 0.000 No configured divergence flag.

Ridge False mae=25808.158; rmse=33312.262; r2=0.493 mae=26603.026 – 60.012; rmse=34393.939 – 158.451; r2=0.476 – 0.005 No configured divergence flag.

Random Forest False mae=19496.268; rmse=25480.222; r2=0.703 mae=20137.235 – 19.265; rmse=26345.693 – 129.358; r2=0.692 – 0.003 Training R² exceeds test R² by 0.255.

HistGradientBoosting False mae=19111.888; rmse=24589.524; r2=0.724 mae=19707.853 – 91.441; rmse=25471.897 – 22.832; r2=0.712 – 0.002 No configured divergence flag.

Evaluation

Selected model: HistGradientBoosting . HistGradientBoosting was selected because it beat the Dummy baseline on held-out r2 by +0.724 and achieved the strongest risk-adjusted score after considering CV stability, train/test divergence, and model complexity.

Class imbalance does not apply to this regression task.

Model Metric Baseline Model Absolute improvement Beats baseline

Ridge r2 -0.000 0.493 0.493 True

Random Forest r2 -0.000 0.703 0.703 True

HistGradientBoosting r2 -0.000 0.724 0.724 True

Held-out permutation importance

Feature Mean Std

V3 0.298 0.017

V1 0.167 0.003

V24 0.075 0.002

V25 0.072 0.005

V11 0.062 0.006

V5 0.043 0.002

V4 0.039 0.003

V22 0.038 0.004

V6 0.028 0.002

V12 0.025 0.001

V26 0.017 0.002

V15 0.016 0.002

V18 0.013 0.003

V19 0.012 0.000

V14 0.012 0.004

V10 0.007 0.001

V9 0.007 0.000

V28 0.005 0.001

V13 0.005 0.000

Amount 0.004 0.001

V16 0.004 0.001

V17 0.003 0.000

V7 0.003 0.000

V21 0.002 0.001

V8 0.002 0.000

V20 0.002 0.000

V23 0.001 0.000

V27 0.001 0.000

V2 0.001 0.000

Class 0.000 0.000

Held-out permutation importance

Held-out permutation importance

Held-out residual diagnostic

Held-out residual diagnostic

Limitations

1,081 duplicate row(s) were detected (0.4% of rows).

'Class' is near-constant: its most common observed value represents 99.8% of non-missing rows.

Categorical distributions: No categorical or boolean columns are present.

Missingness visualization: No missing cells remain after structural cleaning.

Numeric ? categorical relationships: At least one usable numeric and one usable categorical column are required.

Categorical associations: At least two usable categorical columns are required.

Clustering: Best silhouette was 0.080, below the meaningfulness threshold 0.20; separation is too weak to present as segments.

1,081 duplicate row(s) were detected (0.4% of rows). 'Class' is near-constant: its most common observed value represents 99.8% of non-missing rows. Clustering was skipped or rejected: Best silhouette was 0.080, below the meaningfulness threshold 0.20; separation is too weak to present as segments.

Recommendation

What we learned

HistGradientBoosting is the strongest defensible candidate: held-out $r^2=0.724$, improving on baseline by +0.724.

How reliable it appears

The held-out test set contains 4000 rows. 3-fold CV was available. Its CV r^2 was 0.712 – 0.002. No configured train/test divergence threshold was exceeded for the selected model.

What action is reasonable

Use this result as a validated candidate for further business/domain review and prospective testing, not as an automatic production-deployment decision.

What limitations matter

1,081 duplicate row(s) were detected (0.4% of rows). 'Class' is near-constant: its most common observed value represents 99.8% of non-missing rows. Clustering was skipped or rejected: Best silhouette was 0.080, below the meaningfulness threshold 0.20; separation is too weak to present as segments.

What to do next

Review the highest permutation-importance features for plausibility and timing, inspect the held-out error diagnostic, validate performance on fresh data, and only then consider deployment thresholds or business actions.

Technical appendix

Structural rows after preparation 283726

Structural columns after preparation 31

Deferred preprocessing numeric columns Time, V1, V2, V3, V4, V5, V6, V7, V8, V9, V10, V11, V12, V13, V14, V15, V16, V17, V18, V19, V20, V21, V22, V23, V24, V25, V26, V27, V28, Amount, Class

Deferred preprocessing categorical columns

EDA reusable charts 13

Clustering reusable charts 2

Evaluation reusable charts 2

Supervised primary metric r^2